对于一般的二次规划问题

$$\begin{aligned} \min_{x} \quad & f(x)=\frac{1}{2}x^\top Px+c^\top x \\ \text{s.t.} \quad & Gx\le h \\ & Ax=b \end{aligned}$$

之前的推导中引入了$\xi=h-Gx$，并记$H(x)=Ax-b$
改写了$F(x,s;\mu,\rho)$的定义（代入函数Γ（IPRM 中的函数G）的值从x_j改为了ξ_j）：

$$F(x,s;\mu,\rho)=f(x)+\sum_{j=1}^{n}\Gamma(\xi_j,s_j;\mu,\rho)$$

Theorem 2.5 中子问题（17）关联的系统（18）-（20）变为

$$\begin{aligned} \nabla f(x)-\nabla H(x)\lambda+\rho G^\top y&=0 \\ H(x)&=0 \\ z-\xi&=0 \end{aligned}$$

由 Lemma 2.1（1），$z-\xi=0$蕴含$\rho y=s$
把ξ作为松弛变量引入，并且令$\lambda=-t$（让系数与 QOCO 更一致）则系统变为:

$$\begin{aligned} Px+c+A^\top t+G^\top s&=0 \\ Ax&=b \\ Gx+\xi&=h \\ z-\xi&=0 \end{aligned}$$

这样前三式与 QOCO 的(9a)-(9c)对应，(9d)替换为第四式。
因为之前推导时用ξ代替x代入z,y，所以这里

$$z_j=z_j(\xi_j,s_j;\mu,\rho)=\frac{1}{2\rho}\left(\sqrt{\left(s_j-\rho\xi_j\right)^2+4\rho\mu}-\left(s_j-\rho\xi_j\right)\right)$$

$$y_j=y_j(\xi_j,s_j;\mu,\rho)=\frac{1}{2\rho}\left(\sqrt{\left(s_j-\rho\xi_j\right)^2+4\rho\mu}+\left(s_j-\rho\xi_j\right)\right)$$

系统就有(μ,x,t,s,ξ)这五个迭代变量。
定义

$$\phi_{(\mu,\rho)}(x,t,s,\xi)=\frac{1}{2}\left\|\psi_{(\mu,\rho)}(x,t,s,\xi)\right\|^2 \tag{1}$$

$$\psi_{(\mu,\rho)}(x,t,s,\xi)=\begin{bmatrix} Px+c+A^\top t+G^\top s \\ Ax-b \\ z-\xi \\ Gx+\xi-h \end{bmatrix} \tag{2}$$

完整的目标为

$$\mu+\gamma\phi_{(\mu,\rho)}(x,t,s,\xi)=0 \tag{3}$$
$$\psi_{(\mu,\rho)}(x,t,s,\xi)=0 \tag{4}$$

(1)的牛顿方程为

$$\gamma\nabla_{(\mu,x,t,s,\xi)}\phi_{(\mu_k,\rho_k)}(x_k,t_k,s_k,\xi_k)^{\top}\begin{bmatrix}\Delta\mu\\ \mathrm{d_x}\\ \mathrm{d_t}\\ \mathrm{d_s}\\ d_\xi\end{bmatrix}+\Delta\mu=-\mu_k-\gamma\phi_{(\mu_k,\rho_k)}(x_k,t_k,s_k,\xi_k) \tag{5}$$

(2)的牛顿方程为

$$\nabla_{(\mu,x,t,s,\xi)}\psi_{(\mu_k,\rho_k)}(x_k,t_k,s_k,\xi_k)^{\top}\begin{bmatrix}\Delta\mu\\ d_x\\ d_t\\ d_s\\ d_\xi\end{bmatrix}=-\psi_{(\mu_k,\rho_k)}(x_k,t_k,s_k,\xi_k) \tag{6}$$

对(1)求导得

$$\nabla_{(\mu,x,t,s,\xi)}\phi_{(\mu_k,\rho_k)}(x_k,t_k,s_k,\xi_k)=\nabla_{(\mu,x,\lambda,s,\xi)}\psi_{(\mu_k,\rho_k)}(x_k,t_k,s_k,\xi_k)\psi_{(\mu_k,\rho_k)}(x_k,t_k,s_k,\xi_k) \tag{7}$$

将(7)(6)依次代入(5)得

$$-2\gamma\phi_{(\mu_k,\rho_k)}(x_k,t_k,s_k,\xi_k)+\Delta\mu=-\mu_k-\gamma\phi_{(\mu_k,\rho_k)}(x_k,t_k,s_k,\xi_k)$$

得到

$$\Delta\mu=-\mu_k+\gamma\phi_{(\mu_k,\rho_k)}(x_k,t_k,s_k,\xi_k) \tag{8}$$

$$\nabla_{(\mu,x,t,s,\xi)}\psi_{(\mu_k,\rho_k)}(x_k,t_k,s_k,\xi_k)=\begin{bmatrix}O & O & \frac{1}{\rho_k}e^{\top}(Z_k+Y_k)^{-1} & O\\ P & \mathrm{A}^{\top} & O & G^{\top}\\ \mathrm{A} & O & O & O\\ G & O & -\frac{1}{\rho_k}(Z_k+Y_k)^{-1}Z_k & O\\ O & O & -(Z_k+Y_k)^{-1}Y_k & I\end{bmatrix}$$

代入(6):

$$\begin{bmatrix} O & P & A^\top & G^\top & O \\ O & \mathrm{A} & O & O & O \\ \frac{1}{\rho_k}(Z_k+Y_k)^{-1}e & O & O & -\frac{1}{\rho_k}(Z_k+Y_k)^{-1}Z_k & -(Z_k+Y_k)^{-1}Y_k \\ O & G & O & O & I \end{bmatrix}\begin{bmatrix} \Delta\mu \\ d_x \\ d_t \\ d_s \\ d_\xi \end{bmatrix} = \begin{bmatrix} -r_k^d \\ -r_k^h \\ -r_k^e \\ -r_k^\xi \end{bmatrix}$$

$$\begin{bmatrix} P & A^\top & G^\top & O \\ \mathrm{A} & O & O & O \\ O & O & -\frac{1}{\rho_k}(Z_k+Y_k)^{-1}Z_k & -(Z_k+Y_k)^{-1}Y_k \\ G & O & O & I \end{bmatrix}\begin{bmatrix} d_x \\ d_t \\ d_s \\ d_\xi \end{bmatrix} = \begin{bmatrix} -r_k^d \\ -r_k^h \\ -r_k^e - \frac{1}{\rho_k}\Delta\mu(Z_k+Y_k)^{-1}e \\ -r_k^\xi \end{bmatrix}$$

第四行:

$$d_\xi = -r_k^\xi - G d_x \tag{9}$$

用第四行消去第三行的d_ξ:

$$\begin{bmatrix} P & A^\top & G^\top \\ \mathrm{A} & O & O \\ (Z_k+Y_k)^{-1}Y_k G & O & -\frac{1}{\rho_k}(Z_k+Y_k)^{-1}Z_k \end{bmatrix}\begin{bmatrix} d_x \\ d_t \\ d_s \end{bmatrix} = \begin{bmatrix} -r_k^d \\ -r_k^h \\ -r_k^e - \frac{1}{\rho_k}\Delta\mu(Z_k+Y_k)^{-1}e - (Z_k+Y_k)^{-1}Y_k r_k^\xi \end{bmatrix}$$

$$\begin{bmatrix} P & A^\top & G^\top \\ \mathrm{A} & O & O \\ G & O & -\frac{1}{\rho_k}Y_k^{-1}Z_k \end{bmatrix}\begin{bmatrix} d_x \\ d_t \\ d_s \end{bmatrix} = -\begin{bmatrix} r_k^d \\ r_k^h \\ (Z_k+Y_k)Y_k^{-1}r_k^e + \frac{1}{\rho_k}\Delta\mu Y_k^{-1}e + r_k^\xi \end{bmatrix}$$

$$\begin{aligned} r_k^d &= Px_k + c + A^\top t_k + G^\top s_k \\ r_k^h &= Ax_k - b \\ r_k^e &= z_k - \xi_k \\ r_k^\xi &= Gx_k + \xi_k - h \end{aligned}$$

第三行代入$Y_k^{-1} = \frac{\rho_k}{\mu_k}Z_k$

$$\begin{bmatrix} P & A^\top & G^\top \\ \mathrm{A} & O & O \\ G & O & -\frac{1}{\mu_k}Z_k^2 \end{bmatrix}\begin{bmatrix} d_x \\ d_t \\ d_s \end{bmatrix} = -\begin{bmatrix} r_k^d \\ r_k^h \\ (Z_k+Y_k)\frac{\rho_k}{\mu_k}Z_k r_k^e + \frac{\Delta\mu}{\mu_k}Z_k e + r_k^\xi \end{bmatrix}$$

$$\begin{bmatrix} P & A^\top & G^\top \\ \mathrm{A} & O & O \\ \mu_k G & O & -Z_k^2 \end{bmatrix}\begin{bmatrix} d_x \\ d_t \\ d_s \end{bmatrix} = -\begin{bmatrix} r_k^d \\ r_k^h \\ \rho_k(Z_k+Y_k)Z_k r_k^e + \Delta\mu Z_k e + \mu_k r_k^\xi \end{bmatrix}$$

令$d_s = \mu_k \hat{d}_s$

$$\begin{bmatrix} P & A^\top & \mu_k G^\top \\ \mathrm{A} & O & O \\ \mu_k G & O & -\mu_k Z_k^2 \end{bmatrix}\begin{bmatrix} d_x \\ d_t \\ \hat{d}_s \end{bmatrix} = -\begin{bmatrix} r_k^d \\ r_k^h \\ \rho_k(Z_k+Y_k)Z_k r_k^e + \Delta\mu Z_k e + \mu_k r_k^\xi \end{bmatrix}$$

汇总计算公式：

$$z_j = z_j(\xi_j, s_j; \mu, \rho) = \frac{1}{2\rho}\left(\sqrt{\left(s_j - \rho\xi_j\right)^2 + 4\rho\mu} - \left(s_j - \rho\xi_j\right)\right)$$

$$y_j = y_j(\xi_j, s_j; \mu, \rho) = \frac{1}{2\rho}\left(\sqrt{\left(s_j - \rho\xi_j\right)^2 + 4\rho\mu} + \left(s_j - \rho\xi_j\right)\right)$$

$$\psi_{(\mu_k,\rho_k)}(x_k, t_k, s_k, \xi_k) = \begin{bmatrix} r_k^d \\ r_k^h \\ r_k^e \\ r_k^\xi \end{bmatrix} = \begin{bmatrix} Px_k + c + A^\top t_k + G^\top s_k \\ Ax_k - b \\ z_k - \xi_k \\ Gx_k + \xi_k - h \end{bmatrix} \text{ size: } \begin{bmatrix} n \\ p \\ m \\ m \end{bmatrix}$$

$$\phi_{(\mu,\rho)}(x,t,s,\xi) = \frac{1}{2}\left\|\psi_{(\mu,\rho)}(x,t,s,\xi)\right\|^2$$

$$\Delta\mu = -\mu_k + \gamma\phi_{(\mu_k,\rho_k)}(x_k, t_k, s_k, \xi_k)$$

$$\begin{bmatrix} P & A^\top & \mu_k G^\top \\ \mathrm{A} & O & O \\ \mu_k G & O & -\mu_k Z_k^2 \end{bmatrix}\begin{bmatrix} d_x \\ d_t \\ \hat{d}_s \end{bmatrix} = -\begin{bmatrix} r_k^d \\ r_k^h \\ \rho_k(Z_k+Y_k)Z_k r_k^e + \Delta\mu Z_k e + \mu_k r_k^\xi \end{bmatrix}$$

$$d_s = \mu_k \hat{d}_s$$

$$d_\xi = -r_k^\xi - G d_x$$

考虑静态正则化:

在 work->data->P，将P修改为$P + \epsilon_s I$
iprm->construct_kkt 中（2，2）块写成$-\epsilon_s I$
在更新（3,3）块时要加上$-\epsilon_s I$
计算$r_k = \psi_{(\mu_k,\rho_k)}(x_k, t_k, s_k, \xi_k)$时要抵消$P$的静态正则化的影响

求解$d_x, d_t, \hat{d}_s$时要用 iterative refinement 来减小误差:
为求解方程$Kd = r$,

$$\widehat{K}\Delta d^w = r - Kd^w$$

$$d^{w+1} = d^w + \Delta d^w$$

反复用以上两式迭代

$$K_k d = \begin{bmatrix} P & A^\top & \mu_k G^\top \\ \mathrm{A} & O & O \\ \mu_k G & O & -\mu_k Z_k^2 \end{bmatrix} \begin{bmatrix} d_x \\ d_t \\ \hat{d}_s \end{bmatrix} = \begin{bmatrix} Pd_x + A^\top d_t + \mu_k G^\top \hat{d}_s \\ Ad_x \\ \mu_k G d_x - \mu_k Z_k^2 \hat{d}_s \end{bmatrix}$$

当$\mu_k = 1, Z_k = I$时，上面的计算式就特殊化为下式，可以在初始化中用来做 iterative refinement：

$$\begin{bmatrix} P & A^\top & G^\top \\ \mathrm{A} & O & O \\ G & O & -I \end{bmatrix} \begin{bmatrix} x \\ t \\ s \end{bmatrix}$$

IPRM 采用的终止条件缺少相对误差的设计，在此仿照 QOCO 改造一下条件：

$$\left\| \begin{bmatrix} Ax_k - b \\ Gx_k + \xi_k - h \end{bmatrix} \right\|_\infty \le \epsilon_{abs} + \epsilon_{rel} \max\{\|Ax_k\|_\infty, \|b\|_\infty, \|Gx_k\|_\infty, \|h\|_\infty, \|\xi_k\|_\infty\}$$

$$\|Px_k + A^\top t_k + G^\top s_k + c\|_\infty \le \epsilon_{abs} + \epsilon_{rel} \max\{\|Px_k\|_\infty, \|A^\top t_k\|_\infty, \|G^\top s_k\|_\infty, \|c\|_\infty\}$$

$$\|z_k - \xi_k\|_\infty \le \epsilon_{abs} + \epsilon_{rel} \max\{1, |p_k|, |d_k|\}$$

$$p_k \coloneqq \frac{1}{2} x_k^\top P x_k + c^\top x_k, d_k \coloneqq -\frac{1}{2} x_k^\top P x_k - b^\top t_k - h^\top s_k$$

【iprm-3】

$$\begin{bmatrix} P & A^\top & G^\top \\ \mathrm{A} & O & O \\ G & O & -\frac{1}{\rho_k}Y_k^{-1}Z_k \end{bmatrix}\begin{bmatrix} d_x \\ d_t \\ d_s \end{bmatrix} = -\begin{bmatrix} r_k^d \\ r_k^h \\ (Z_k+Y_k)Y_k^{-1}r_k^e + \frac{1}{\rho_k}\Delta\mu Y_k^{-1}e + r_k^\xi \end{bmatrix}$$

第三行代入$Y_k^{-1} = \frac{\rho_k}{\mu_k}Z_k$

$$\begin{bmatrix} P & A^\top & G^\top \\ \mathrm{A} & O & O \\ G & O & -\frac{1}{\mu_k}Z_k^2 \end{bmatrix}\begin{bmatrix} d_x \\ d_t \\ d_s \end{bmatrix} = -\begin{bmatrix} r_k^d \\ r_k^h \\ (Z_k+Y_k)\frac{\rho_k}{\mu_k}Z_k r_k^e + \frac{\Delta\mu}{\mu_k}Z_k e + r_k^\xi \end{bmatrix}$$

第三行乘以$\sqrt{\mu_k}$

$$\begin{bmatrix} P & A^\top & G^\top \\ \mathrm{A} & O & O \\ \sqrt{\mu_k}G & O & -\frac{1}{\sqrt{\mu_k}}Z_k^2 \end{bmatrix}\begin{bmatrix} d_x \\ d_t \\ d_s \end{bmatrix} = -\begin{bmatrix} r_k^d \\ r_k^h \\ (Z_k+Y_k)\frac{\rho_k}{\sqrt{\mu_k}}Z_k r_k^e + \frac{\Delta\mu}{\sqrt{\mu_k}}Z_k e + \sqrt{\mu_k}r_k^\xi \end{bmatrix}$$

令$d_s = \sqrt{\mu_k}\hat{d}_s$

$$\begin{bmatrix} P & A^\top & \sqrt{\mu_k}G^\top \\ \mathrm{A} & O & O \\ \sqrt{\mu_k}G & O & -Z_k^2 \end{bmatrix}\begin{bmatrix} d_x \\ d_t \\ \hat{d}_s \end{bmatrix} = -\begin{bmatrix} r_k^d \\ r_k^h \\ (Z_k+Y_k)\frac{\rho_k}{\sqrt{\mu_k}}Z_k r_k^e + \frac{\Delta\mu}{\sqrt{\mu_k}}Z_k e + \sqrt{\mu_k}r_k^\xi \end{bmatrix}$$

【iprm-2】

$$\begin{bmatrix} P & A^\top & G^\top \\ \mathrm{A} & O & O \\ G & O & -\frac{1}{\rho_k}Y_k^{-1}Z_k \end{bmatrix}\begin{bmatrix} d_x \\ d_t \\ d_s \end{bmatrix} = -\begin{bmatrix} r_k^d \\ r_k^h \\ (Z_k+Y_k)Y_k^{-1}r_k^e + \frac{1}{\rho_k}\Delta\mu Y_k^{-1}e + r_k^\xi \end{bmatrix}$$

$$\begin{aligned} r_k^d &= Px_k + c + A^\top t_k + G^\top s_k \\ r_k^h &= Ax_k - b \\ r_k^e &= z_k - \xi_k \\ r_k^\xi &= Gx_k + \xi_k - h \end{aligned}$$

第三行代入$Y_k^{-1} = \frac{\rho_k}{\mu_k}Z_k$

$$\begin{bmatrix} P & A^\top & G^\top \\ \mathrm{A} & O & O \\ G & O & -\frac{1}{\mu_k}Z_k^2 \end{bmatrix}\begin{bmatrix} d_x \\ d_t \\ d_s \end{bmatrix} = -\begin{bmatrix} r_k^d \\ r_k^h \\ (Z_k+Y_k)\frac{\rho_k}{\mu_k}Z_k r_k^e + \frac{\Delta\mu}{\mu_k}Z_k e + r_k^\xi \end{bmatrix}$$

$$\begin{bmatrix} P & A^\top & G^\top \\ \mathrm{A} & O & O \\ G & O & -\frac{1}{\mu_k}Z_k^2 \end{bmatrix}\begin{bmatrix} x \\ t \\ s \end{bmatrix} = \begin{bmatrix} Px + A^\top t + G^\top s \\ Ax \\ Gx - \frac{1}{\mu_k}Z_k^2 s \end{bmatrix}$$

【iprm-1】

$$\begin{bmatrix} P & A^\top & G^\top \\ \mathrm{A} & O & O \\ \sqrt{\mu_k}Z_k^{-1}G & O & -\frac{1}{\sqrt{\mu_k}}Z_k \end{bmatrix}\begin{bmatrix} d_x \\ d_t \\ d_s \end{bmatrix} = -\begin{bmatrix} r_k^d \\ r_k^h \\ (Z_k+Y_k)\frac{\rho_k}{\sqrt{\mu_k}}r_k^e + \frac{\Delta\mu}{\sqrt{\mu_k}}e + \sqrt{\mu_k}Z_k^{-1}r_k^\xi \end{bmatrix}$$

令$d_s = \sqrt{\mu_k}Z_k^{-1}\hat{d}_s$

$$\begin{bmatrix} P & A^\top & \sqrt{\mu_k}G^\top Z_k^{-1} \\ \mathrm{A} & O & O \\ \sqrt{\mu_k}Z_k^{-1}G & O & -I \end{bmatrix}\begin{bmatrix} d_x \\ d_t \\ \hat{d}_s \end{bmatrix} = -\begin{bmatrix} r_k^d \\ r_k^h \\ (Z_k+Y_k)\frac{\rho_k}{\sqrt{\mu_k}}r_k^e + \frac{\Delta\mu}{\sqrt{\mu_k}}e + \sqrt{\mu_k}Z_k^{-1}r_k^\xi \end{bmatrix}$$

用于 iterative refinement：

$$\begin{bmatrix} P & A^\top & \sqrt{\mu_k}G^\top Z_k^{-1} \\ \mathrm{A} & O & O \\ \sqrt{\mu_k}Z_k^{-1}G & O & -I \end{bmatrix}\begin{bmatrix} x \\ t \\ s \end{bmatrix} = \begin{bmatrix} Px + A^\top t + \sqrt{\mu_k}G^\top Z_k^{-1}s \\ Ax \\ \sqrt{\mu_k}Z_k^{-1}Gx - s \end{bmatrix}$$